2S servo power / Radxa ZERO 3W / BMI088 + ToF / audio

Excluded: RS485, identification EEPROM, charging, UPS, camera and display. Not released for fabrication.

radxa_40p

文件: radxa_40p.kicad_sch

power

文件: power.kicad_sch

servo_ttl

文件: servo_ttl.kicad_sch

imu_tof

文件: imu_tof.kicad_sch

audio

文件: audio.kicad_sch

testpoints

文件: testpoints.kicad_sch

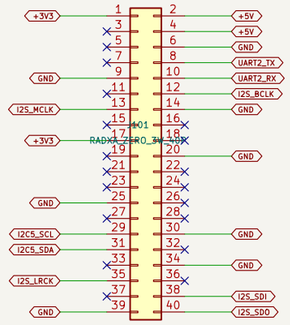
duck-robot

Sheet: /
File: custom_hat_rev_a.kicad_sch

Title: Custom HAT Rev A – functional schematic

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 1/7

3.3V GPIO only. Pin 8/10 UART2 also used by boot console: disable console before servo bus operation.



I2C5: pins 29/31. I2S3: 12/13/35/38/40. SPI3: 19/21/23/24. No EEPROM.

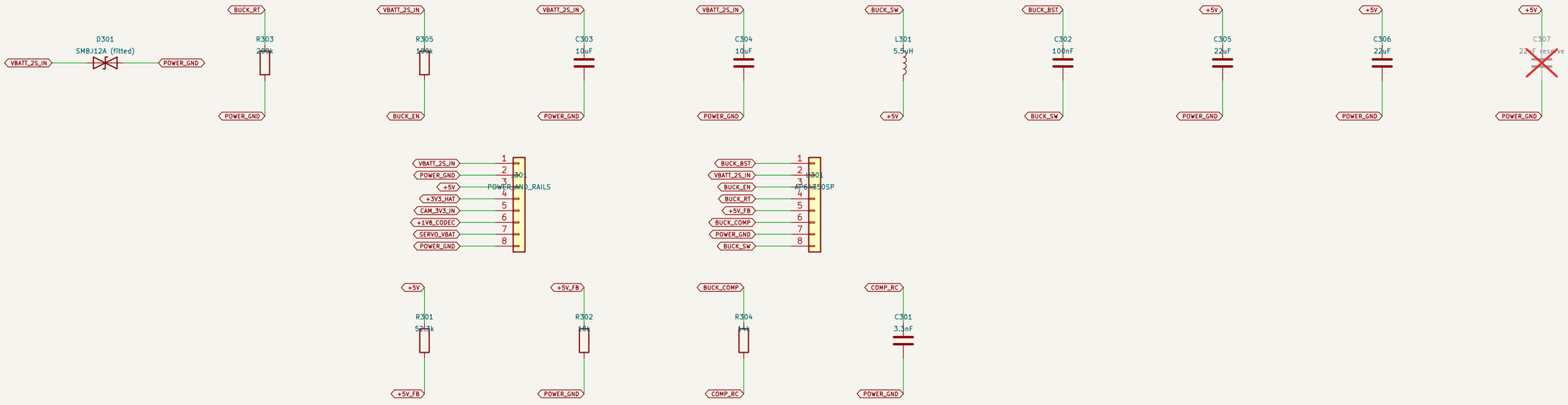
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Sheet: /radxa_40p/
File: radxa_40p.kicad_sch

Title: Radxa ZERO 3W – 40-pin interface / 3.3V logic

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 2/7

AP64350 pin map: 1 BST, 2 VIN, 3 EN, 4 RT/CLK, 5 FB, 6 COMP, 7 GND, 8 SW; EP=GND.



Vout=0.8V*(1+52.3k/10k)=4.98V; fsw=500kHz via RT=200k; C305+C306 fitted (44uF), C307 22uF reserved DNP.

SMBJ12A fitted: hot-plug sweep peaks 14-16V over 1-10uH cable, below the 35V abs-max; TVS is defensive.

Pin 4 is RT/CLK here but SS on AP64501: symbols and PCB must not be reused across the two parts.

duck-robot		
Sheet: /power/		
File: power.kicad_sch		
Title: Power input and AP64350 5V buck - values from DS41976 and EVM B		
Size: A3	Date: 2026-10-28	Rev: A schematic draft
KiCad E.D.A. 10.0.7		Id: 3/7

TTL-only shared DATA path with automatic TX direction control. No RS485 transceiver and no RX AND-gate.







UART2 console conflict remains a software bring-up constraint; do not assume console disabled.

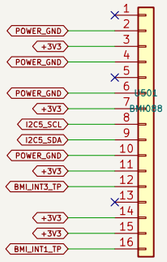
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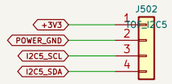
Sheet: /servo_ttl/
File: servo_ttl.kicad_sch

Title: Half-duplex TTL servo bus — two 3P outlets

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 4/7

BMI088 Rev1.9 contract: accel SDO1=3V3 (0x19), gyro SDO2=GND (0x68), PS=3V3. Pin 2 is GND despite NC name.





INT3/INT1 are testpoint/candidate GPIO nets; exact host GPIO assignment remains pending.

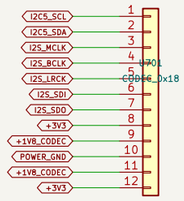
duck-robot

Sheet: /imu_tof/
File: imu_tof.kicad_sch

Title: BMI088 + ToF on shared I2C5

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 5/7

Codec address 0x18. MCLK pin 1 goes to Radxa pin 13; codec DVDD pin 32 requires 1.8V; IO VDD pin 7 is 3.3V.



This page is a pin contract pending full C1 audio schematic re-implementation; no fabricated analog values are implied.

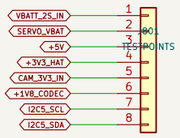
duck-robot

Sheet: /audio/
File: audio.kicad_sch

Title: I2S codec interface — I2C5 and 1V8 codec rail

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 6/7

Test points are measurement access only; no production load or protective function.





duck-robot

Sheet: /testpoints/
File: testpoints.kicad_sch

Title: Bring-up test points

Size: A3	Date: 2026-10-28	**Rev: A schematic draft**
KiCad E.D.A. 10.0.7		Id: 7/7